IGCSE 0607 Practice Questions — E7.4

Trigonometric functions

Total Questions: 21 — **Total Marks:** 131 Marks

Table of Contents & Quick Navigation

- **E7.4** Trigonometric functions
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

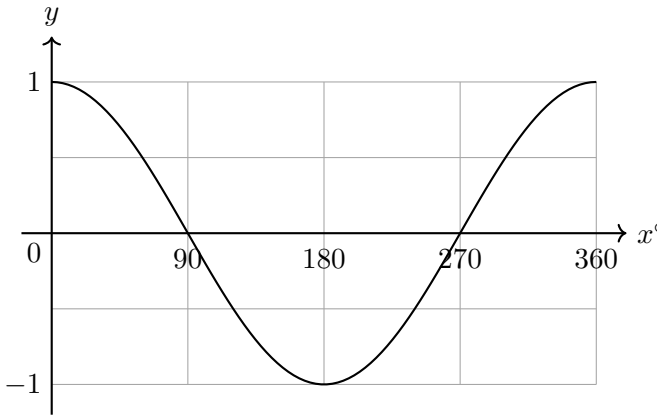
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1 Solve the equation for $0° \leq x \leq 360°$.

$\sqrt{2}\sin x - 1 = 0$

$x =$ ..... **[3]**

2 The diagram shows the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



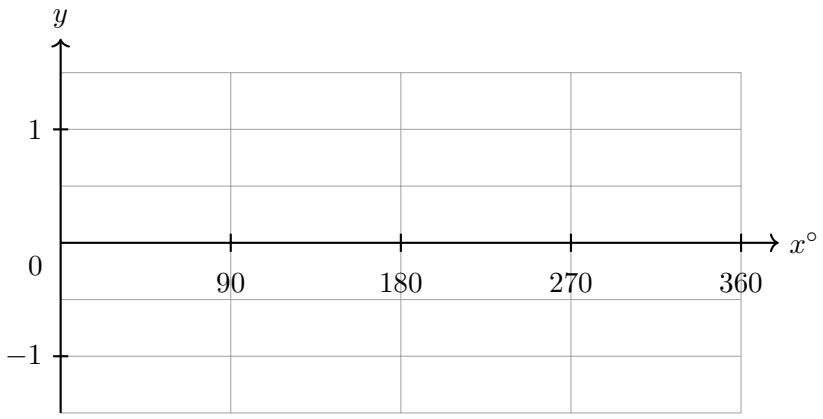
(a) Write down the coordinates of the lowest point on the graph.

(..... , ...) **[1]**

(b) Use the graph, or otherwise, to solve the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

3 **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the exact solutions to the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

4 Answer the following questions for $0° \leq x \leq 360°$.

(a) Solve the equation

$$2\cos x - 1 = 0$$

$x =$ ..... **[3]**

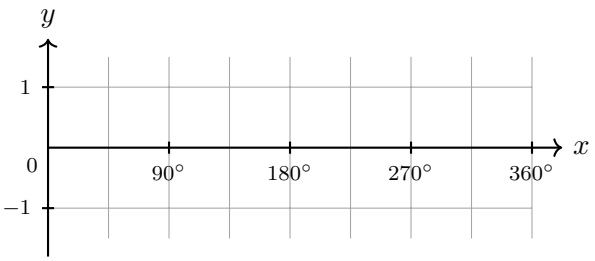
(b) Write down the number of solutions to the equation

$$3\sin x = 5$$

..... **[1]**

5 Consider the trigonometric function $y = \cos x$ for $0° \leq x \leq 360°$.

(a) On the grid below, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



[2]

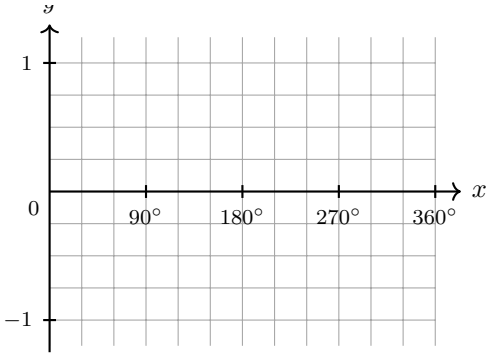
(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\cos x + 1 = 0$$

$x =$ ..... , $x =$ ..... **[3]**

6 Do not use a calculator in this question.

(a) On the axes below, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the coordinates of the maximum point and the minimum point of $y = \sin x$ for $0° \leq x \leq 360°$.

Maximum = ... , Minimum = ... **[2]**

(c) Solve the equation for $0° \leq x \leq 360°$:

(i) $\sin x = \dfrac{\sqrt{3}}{2}$

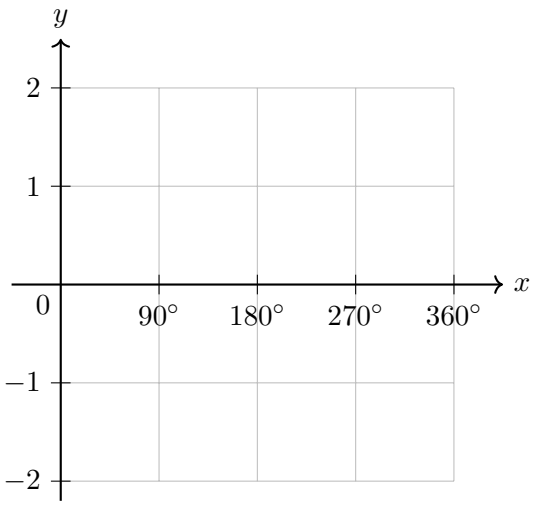
$x =$ ..... , $x =$ ..... **[2]**

(ii) $2\cos x + 1 = 0$

$x =$ ..... , $x =$ ..... **[3]**

7 The properties of trigonometric graphs can be used to solve equations.

(a) On the grid, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the coordinates of the minimum point on this graph.

(..... , ...) **[1]**

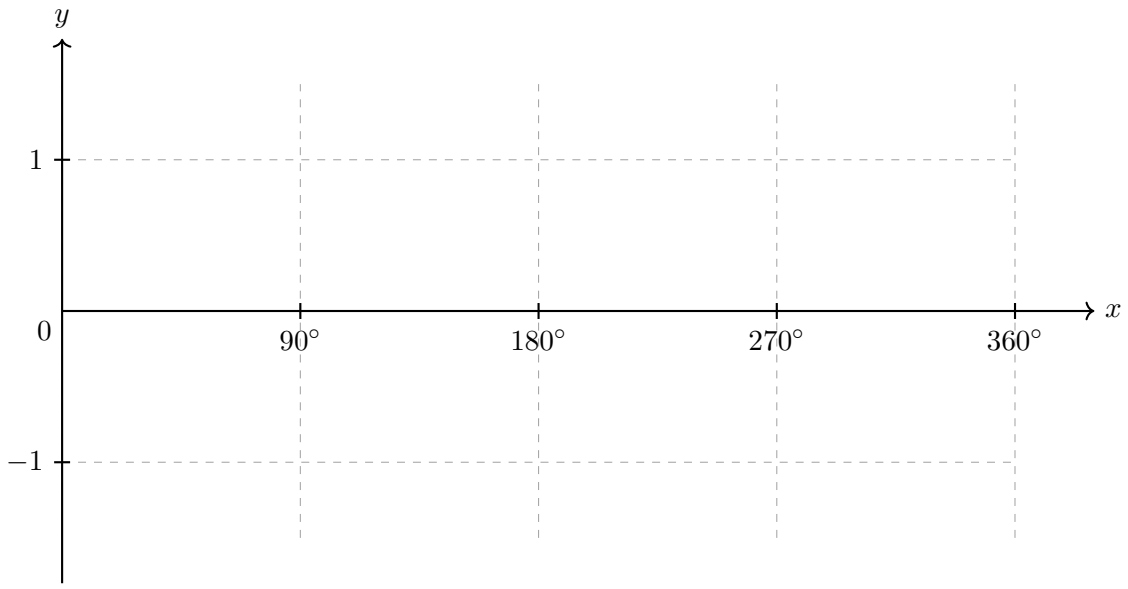
(c) Solve the equation for $0° \leq x \leq 360°$.

$2\cos x + 1 = 0$

$x =$ ..... **[3]**

8 The functions $g(x) = \sin x$ and $h(x) = \cos x$ are defined for $0° \leq x \leq 360°$.

(a) On the grid provided, sketch the graphs of $y = g(x)$ and $y = h(x)$.



[3]

(b) Solve the equation $g(x) = h(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

(c) Use your graphs to write down the range of values of x for which $g(x) > h(x)$ for $0° \leq x \leq 360°$.

..... **[2]**

9 Solve the following equations for $0° \leq x \leq 360°$.

(a) $2\sin x + \sqrt{3} = 0$

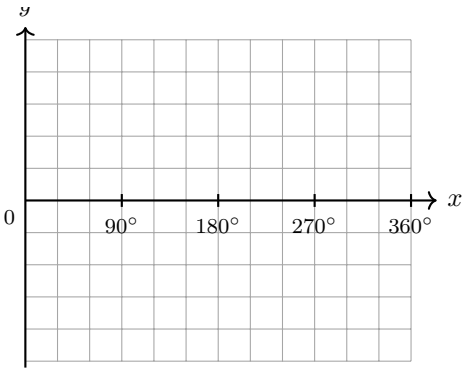
$x =$ ..... **[3]**

(b) $4\cos^2 x - 3 = 0$

$x =$ ..... **[4]**

10 Do not use a calculator in this question.

(a) On the axes below, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.
Draw dashed lines to show any asymptotes.



[3]

(b) Write down the equations of the two asymptotes of the graph of $y = \tan x$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(c) Solve the equation for $0° \leq x \leq 360°$:

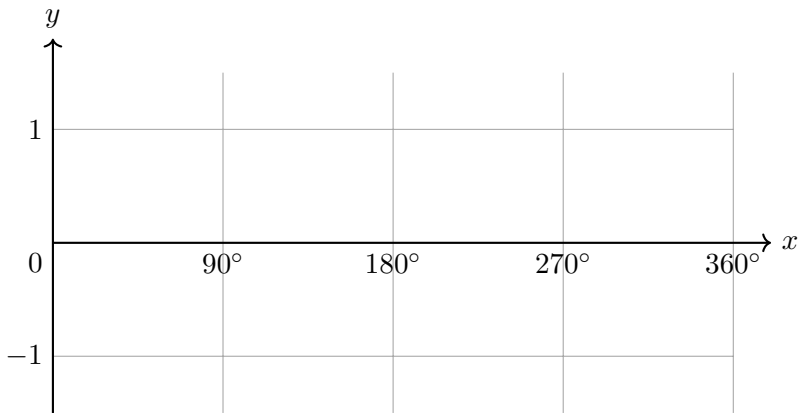
(i) $\sqrt{3}\tan x + 1 = 0$

$x =$ ..... , $x =$ ..... **[3]**

(ii) $2\sin^2 x - \sin x - 1 = 0$

$x =$ ..... , $x =$ ..... , $x =$ ..... **[4]**

11 **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Solve the equation

$\sin x = 0.7$ for $0° \leq x \leq 360°$.

$x =$ ..... **[3]**

12 Consider the trigonometric function $f(x) = 4\sin x + 3$ for $0° \leq x \leq 360°$.

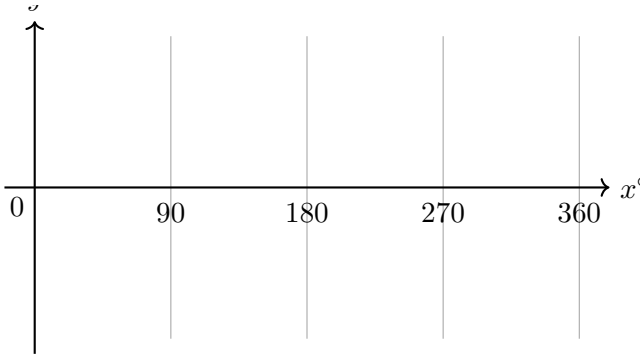
(a) Solve the equation $f(x) = 0$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Write down the exact coordinates of the minimum point on the graph of $y = f(x)$.

(..... , ...) **[2]**

13 **(a)** On the axes provided, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.



[2]

(b) Solve the equation $3\tan x + 5 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

14 **(a)** Solve the equation for $0° \leq x \leq 360°$.

$4\cos x + 1 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

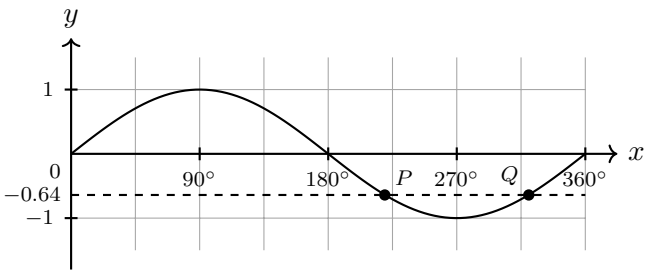
(b) Solve the equation for $0° \leq x \leq 360°$.

$2\tan x - 5 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

15 The diagram shows the graph of $y = \sin x$ for $0° \leq x \leq 360°$.
The line $y = -0.64$ intersects the curve at points P and Q.



(a) Calculate the x-coordinates of P and Q.
Give your answers correct to 1 decimal place.

x-coordinate of $P =$ ..... , x-coordinate of $Q =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$3 \tan x + 5 = 0$ Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

16 **(a)** Solve the equation

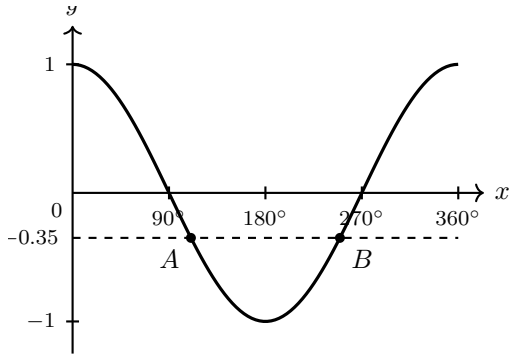
$3\cos x + 1 = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... **[4]**

(b) The graph of $y = 3\cos x + 1$ crosses the x-axis at points A and B.
Calculate the distance between A and B.

Distance = ... **[3]**

17 The diagram shows the graph of $y = \cos x$ and the line $y = -0.35$ for $0° \leq x \leq 360°$.
The line intersects the curve at points A and B.



(a) Calculate the x-coordinate of A and the x-coordinate of B, giving your answers correct to 1 decimal place.

x-coordinate of $A =$ ..... , x-coordinate of $B =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$, giving your answers correct to 1 decimal place:

$5 \sin x + 3 = 0$

$x =$ ..... , $x =$ ..... **[3]**

(c) Solve the equation for $0° \leq x \leq 360°$, giving your answers correct to 1 decimal place:

$3 \tan x - 7 = 0$

$x =$ ..... , $x =$ ..... **[3]**

18 **(a)** The equation $a \sin x - b = 0$ has a solution $x = 30°$.
Given that a and b are positive integers with no common factors, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the second solution to the equation $a \sin x - b = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

(c) A horizontal line $y = k$ intersects the graph of $y = \cos x$ at exactly two points in the interval $0° \leq x \leq 360°$.
The x-coordinate of one of these points is $242°$.
Find the value of k correct to 3 significant figures and find the x-coordinate of the second point of intersection.

$k =$ ..... , $x =$ ..... **[3]**

19 Algebraic techniques can be applied to solve trigonometric equations.

(a) Solve the equation for $0° \leq x \leq 360°$.

$$5\tan x + 14 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\sin^2 x - \sin x = 0$$

$x =$ ..... **[4]**

20 Solve the equation

$$5\sin x = 2\tan x \text{ for } 0° \leq x \leq 360°.$$

Show all your working.

$x =$ ..... **[7]**

21 Solve the following trigonometric equations for $0° \leq x \leq 360°$.

(a)

$5\sin x - 2\cos x = 0$ Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

(b)

$2\sin^2 x + \sin x - 1 = 0$

$x =$ .....

$x =$ .....

$x =$ ..... **[5]**